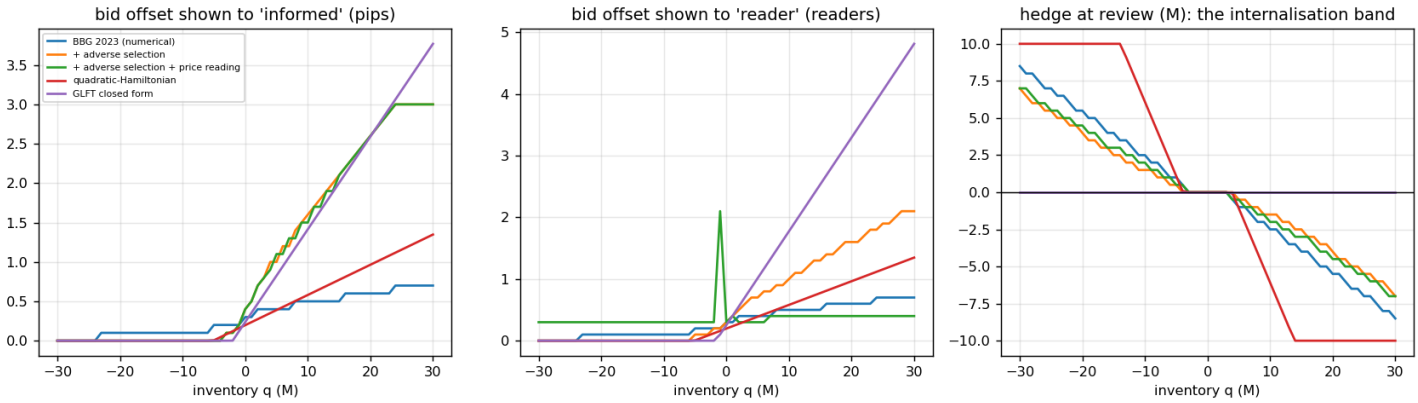
FX Dealer Market Making: Skew, Internalisation, Adverse Selection and Pri

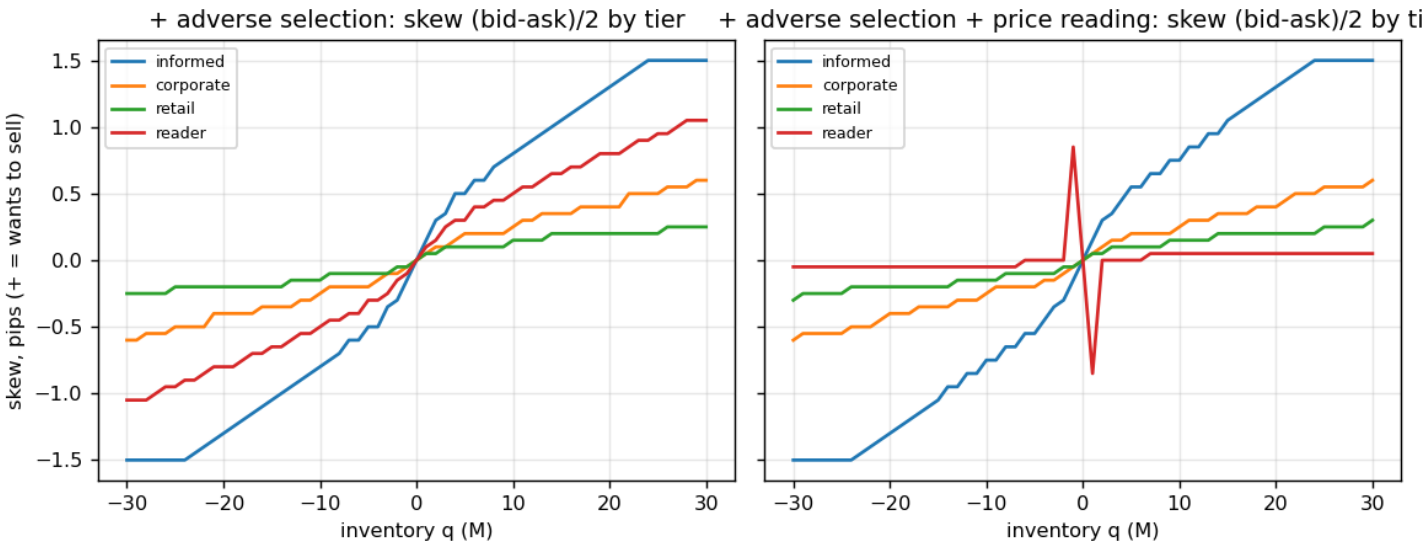
Question. How should a dealer skew and hedge when some clients are informed, some read the dealer's skew, and hedging on the interbank venue has impact?

Method. A C++ event-driven dealer simulator: reference price from EURUSD ticks (histdata.com, April 2025), client requests as marked point processes on a tiered pricing ladder with logistic spread response, tier-specific post-trade drift (adverse selection), skew readers, and hedging with temporary and permanent impact. Quoters: Avellaneda-Stoikov (naive), the GLFT 2013 closed form (sanity), a quadratic-Hamiltonian closed form with adverse selection, and the Barzykin-Bergault-Gueant 2023 dealer solved numerically (ergodic HJB on an inventory grid) with the 2025 adverse-selection / price-reading extension. On top: an online Bayesian mark-out classifier routing toxic trades to the hedge (Cartea & Sanchez-Betancourt 2025), anticipatory hedging of autocorrelated flow (Nutz-Webster-Zhao 2025) and last look as a control. P&L is decomposed into spread capture, inventory, adverse selection, hedging execution and hedging impact, exactly.

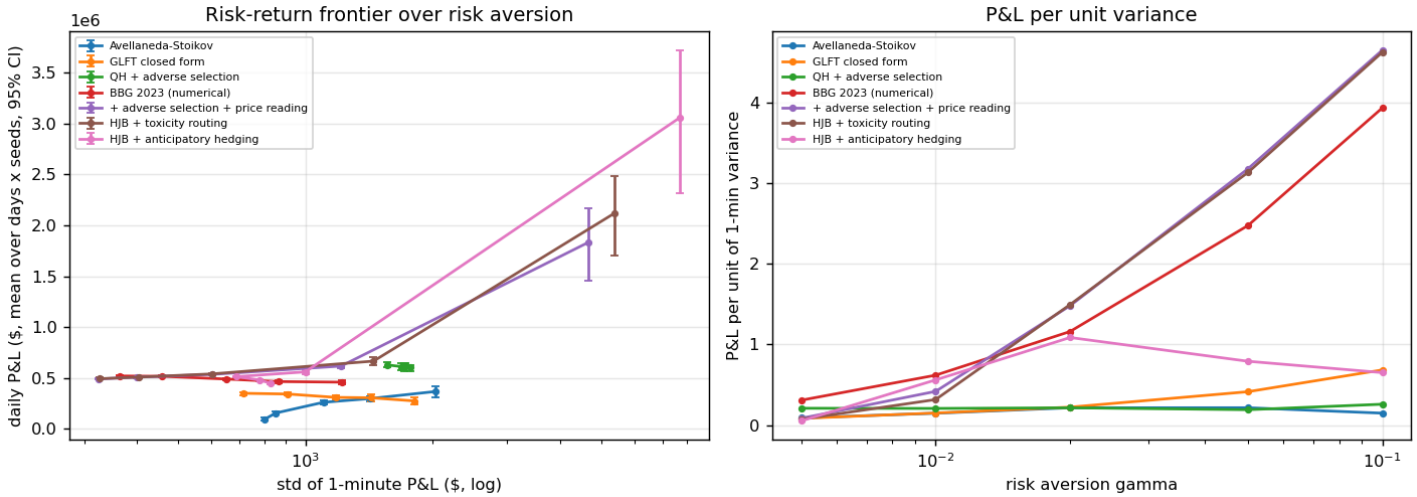
Caveats. Real single-dealer client flow does not exist publicly: the client layer is a calibrated simulation (BIS 2025 turnover shares, Butz-Oomen internalisation horizons, Oomen / LMAX mark-out shapes) and the contribution is reproducing the 2021-25 closed forms and testing the policies on the risk-return frontier.



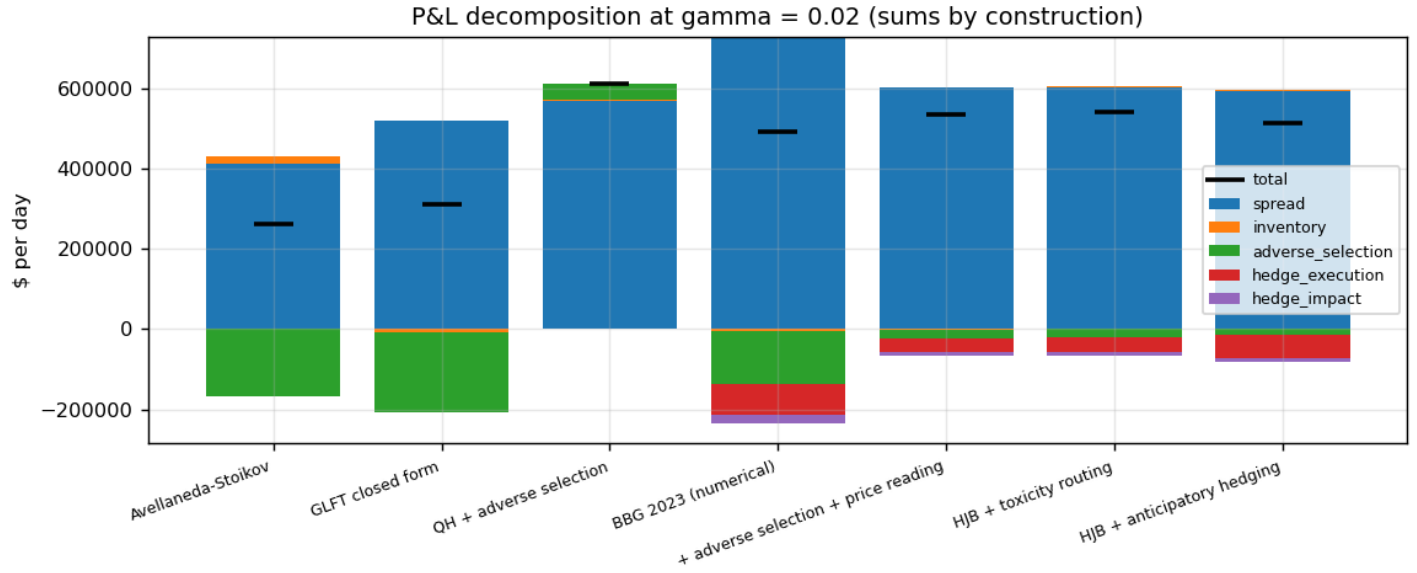
Ladders over inventory: offsets shown to the informed tier and to readers, and the hedging band.



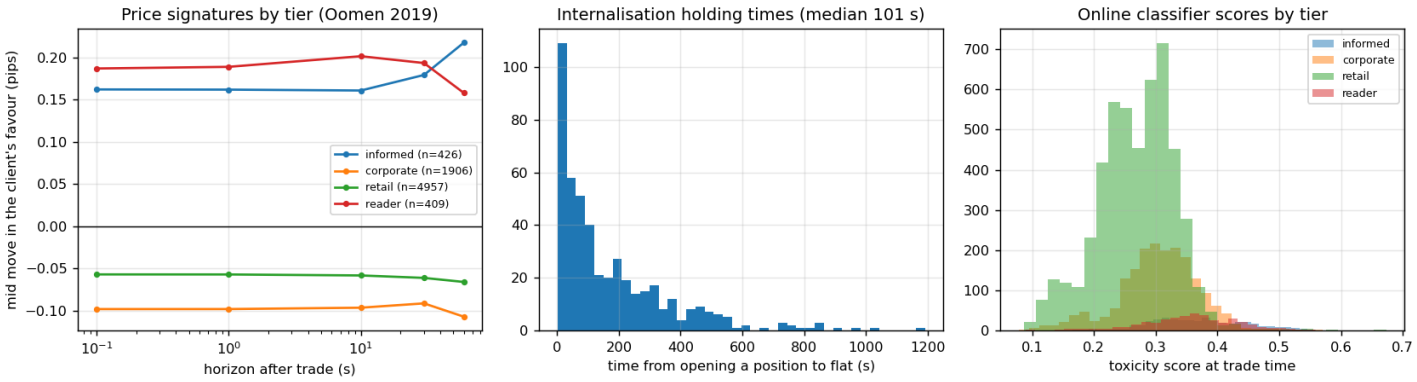
Skew by tier without and with price reading: readers are shown a flat ladder.



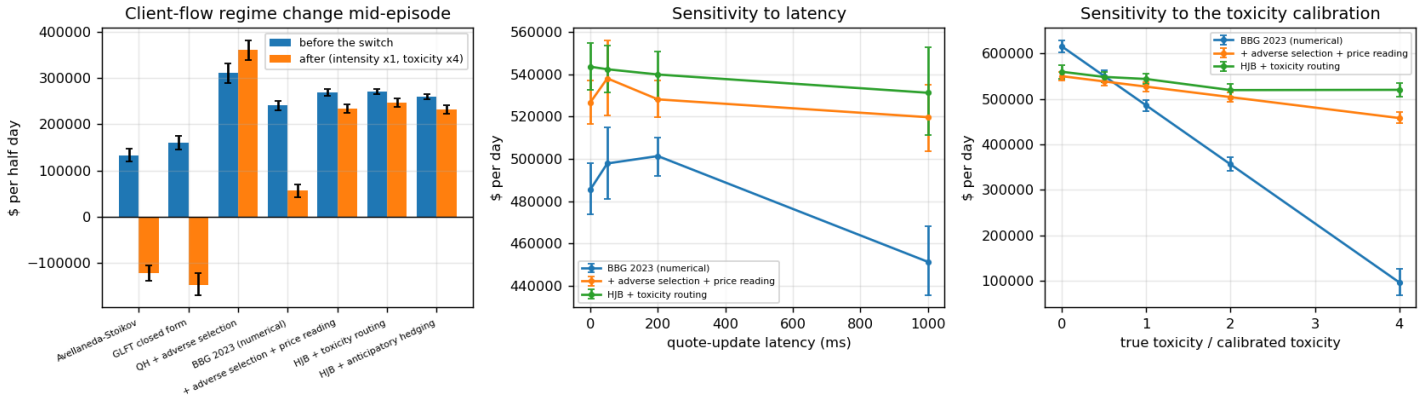
Risk-return frontier over risk aversion, with 95% bootstrap bands over days x seeds.



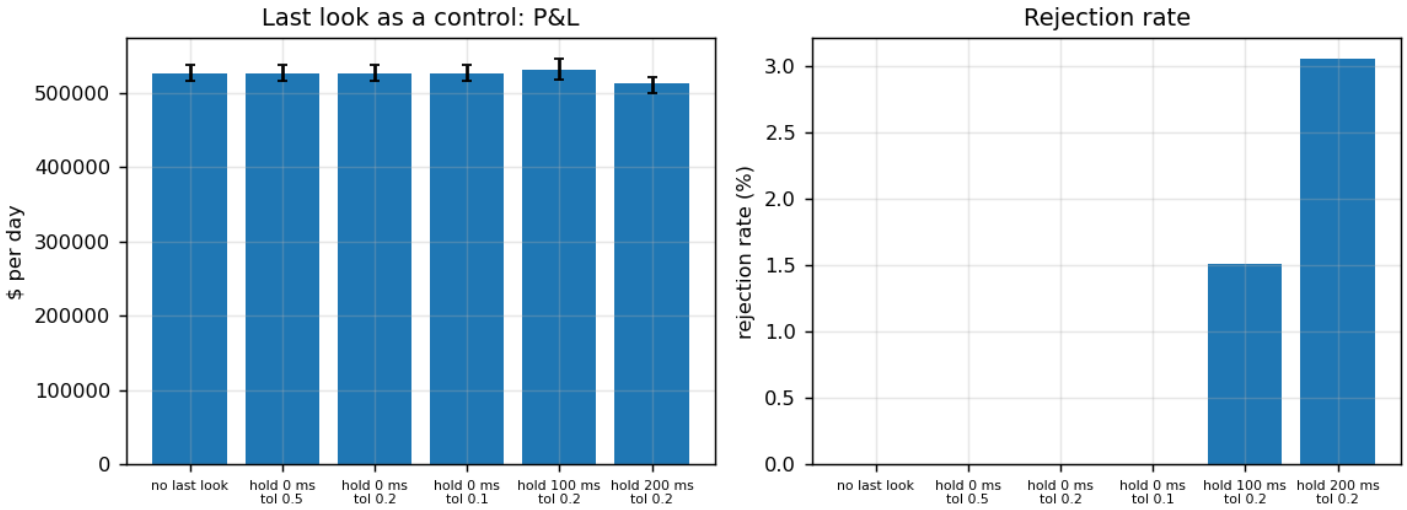
P&L decomposition by quoter (sums by construction).



Price signatures by tier, internalisation holding times and classifier scores.



Client-flow regime change, latency sensitivity and toxicity-calibration sensitivity.



Last look as a control: P&L and rejection rate by hold time and tolerance.

Reference price data (histdata.com EURUSD ticks, April 2025)

day	ticks	median spread (pips)	sigma 1 s (pips/sqrt s)	sigma from 1-min (pips/sqrt s)
2025-04-14	184,392	0.60	0.335	0.340
2025-04-15	144,259	0.60	0.268	0.281
2025-04-16	162,680	0.60	0.290	0.292
2025-04-17	135,613	0.60	0.278	0.256
2025-04-18 (Good Friday, excluded)	164,168	1.30	0.167	0.117

Quoting policies (sigma = 0.30 pips/sqrt s, gamma = 0.02, psi = 0.15 pips/M)

quoter	offsets at q = 0 by tier (pips)	skew slope by tier (pips per M, from q =	hedge band (M)
AS	0.36/0.36/0.36/0.36	0.072/0.072/0.072/0.072	0.0
GLFT	0.24/0.36/0.53/0.25	0.071/0.079/0.108/0.089	0.0
QH	0.20/0.32/0.51/0.20	0.029/0.035/0.038/0.029	3.9
QH_adverse	0.30/0.32/0.50/0.24	0.075/0.020/0.000/0.042	150000000.0
HJB_2023	0.30/0.50/0.70/0.30	0.020/0.015/0.015/0.020	3.0
HJB_noread	0.40/0.50/0.70/0.30	0.080/0.025/0.015/0.050	4.0
HJB	0.40/0.50/0.70/0.30	0.075/0.025/0.015/0.005	3.0

tiers: informed, corporate, retail, reader

mbt_gym check: k = 3.285, gamma = 0.02, sigma = 0.3, H = 60.0: max |fxmm AS - mbt_gym AS| over q in [-Q, Q] = 4.60e-10 pips; PASS

Frontier (4 days x 4 seeds per cell)

quoter	gamma	daily P&L $ [95% CI]	P&L / variance	internalisation	holding time (s)	ema50 (s)	q	(M)	spread	inventory	adverse	hedge exec	hedge impact
AS	0.005	367,643 [312,028, 417,405]	0.089	1.00	184	29.2	537,488	-10,781	-159,064	0	0		
AS	0.01	298,469 [273,423, 322,933]	0.147	1.00	149	22.1	481,817	-11,531	-171,817	0	0		
AS	0.02	261,650 [238,051, 282,676]	0.215	1.00	109	17.1	411,872	17,054	-167,277	0	0		
AS	0.05	156,391 [140,853, 173,775]	0.216	1.00	76	12.5	308,222	8,796	-160,627	0	0		
AS	0.1	95,187 [78,788, 111,066]	0.148	1.00	56	10.8	242,610	3,347	-150,770	0	0		
GLFT	0.005	275,211 [239,942, 308,364]	0.084	1.00	188	28.6	574,337	-12,657	-286,469	0	0		
GLFT	0.01	306,299 [272,428, 338,007]	0.150	1.00	151	23.8	545,461	840	-240,002	0	0		
GLFT	0.02	310,113 [287,230, 332,928]	0.223	1.00	111	19.7	518,651	-8,894	-199,643	0	0		
GLFT	0.05	343,061 [328,168, 357,840]	0.414	1.00	78	16.4	477,157	256	-134,353	0	0		
GLFT	0.1	350,276 [334,465, 364,706]	0.685	1.00	60	14.1	442,620	-342	-92,001	0	0		
QH_adverse	0.005	621,433 [593,123, 649,336]	0.209	1.00	147	30.0	566,692	12,113	42,628	0	0		
QH_adverse	0.01	607,373 [571,533, 641,653]	0.207	1.00	139	30.2	567,611	-6,365	46,127	0	0		
QH_adverse	0.02	610,078 [574,646, 645,528]	0.215	1.00	147	29.8	566,883	3,050	40,145	0	0		
QH_adverse	0.05	602,085 [569,745, 631,448]	0.191	1.00	133	30.2	567,301	-8,927	43,710	0	0		
QH_adverse	0.1	629,469 [605,552, 653,228]	0.259	1.00	148	26.3	567,734	30,932	32,110	-812	-495		
HJB_2023	0.005	459,557 [437,220, 481,933]	0.309	0.84	222	17.2	726,468	13,578	-219,141	-39,453	-21,895		
HJB_2023	0.01	465,429 [450,674, 480,539]	0.619	0.78	153	15.2	725,219	-6,548	-174,987	-56,248	-22,008		
HJB_2023	0.02	490,427 [479,530, 502,094]	1.161	0.69	127	13.5	726,275	-6,053	-129,670	-78,050	-22,075		
HJB_2023	0.05	517,539 [510,457, 524,527]	2.475	0.58	89	12.0	720,178	-2,427	-74,291	-106,194	-19,726		
HJB_2023	0.1	520,481 [513,368, 528,545]	4.935	0.47	43	11.3	714,182	1,498	-44,891	-133,153	-17,155		
HJB	0.005	1,831,489 [1,496,188, 2,166,732]	0.732	0.54	0	34.2	530,582	19,463	1,241,535	-122,576	162,484		
HJB	0.01	617,480 [594,289, 640,710]	0.418	0.98	179	17.2	614,018	14,766	-5,527	-3,674	-2,103		
HJB	0.02	535,784 [524,046, 548,050]	0.481	0.82	125	12.2	601,442	-1,955	-22,940	-31,588	-9,176		
HJB	0.05	508,239 [498,005, 518,730]	0.679	0.66	86	11.0	590,803	-4,534	-12,147	-56,568	-9,315		
HJB	0.1	491,095 [484,279, 497,856]	0.651	0.56	41	10.4	578,609	31	-7,312	-72,164	-8,069		
HJB_tox	0.005	2,119,588 [1,785,389, 2,476,730]	0.847	0.47	0	37.9	504,774	-305	1,607,482	-209,890	217,526		
HJB_tox	0.01	666,434 [628,952, 704,890]	0.517	0.90	170	17.5	605,992	9,966	74,171	-29,796	6,101		
HJB_tox	0.02	540,577 [531,708, 550,031]	0.495	0.80	122	12.6	602,517	3,037	-21,123	-34,775	-9,079		
HJB_tox	0.05	509,161 [500,891, 517,903]	0.733	0.66	85	10.4	589,671	-1,739	-11,663	-58,063	-9,046		
HJB_tox	0.1	493,576 [484,270, 503,442]	0.625	0.56	41	10.7	579,688	2,251	-7,078	-73,323	-7,963		
HJB_nwz	0.005	3,058,152 [2,736,169, 3,372,245]	1.205	0.50	13584	52.3	517,473	98,972	2,272,887	-134,886	303,706		
HJB_nwz	0.01	560,782 [541,956, 577,864]	0.561	0.95	131	14.0	604,444	-6,600	-23,730	-9,590	-3,742		
HJB_nwz	0.02	513,706 [505,847, 522,471]	0.487	0.66	59	12.5	591,183	4,941	-14,628	-59,399	-8,392		
HJB_nwz	0.05	480,467 [470,791, 491,170]	0.792	0.40	39	14.3	581,043	1,122	6,789	-103,214	-5,274		

HJB_nwz	0.1	448,174 [434,898, 462,540]	0.51	0.23	27	14.3	577,322	-5,577	13,417	-134,670	-2,318

One day of the full model (HJB with adverse selection and price reading, 2025-04-14, seed 3)

- P&L $561,100 = spread 609,855 + inventory 21,090 + adverse selection -18,566 + hedge execution -41,163 + hedge impact -10,116 (residual 2.4e-10)
- 7,702 client trades (10,301 M), 3,595 hedges (2,434 M), internalisation ratio 0.76, holding time p50 101 s / mean 173 s, |q| mean 1.7 M max 12.0 M

							tier
informed	426	0.26	+0.16	+0.16	+0.16	+0.18	+0.22
corporate	1,906	0.49	-0.10	-0.10	-0.10	-0.09	-0.11
retail	4,957	0.68	-0.06	-0.06	-0.06	-0.06	-0.07
reader	409	0.39	+0.19	+0.19	+0.20	+0.19	+0.16

classifier: 7,698 labelled trades, 2,142 toxic labels; weights ['0.18', '-0.51', '-0.80', '-0.33', '-1.46', '0.21', '0.50', '-0.20']

Regime change at mid-day (client intensity x1, toxicity x4)

quoter	P&L before ($, 95% CI)	P&L after	survives (CI > 0)
AS	132,954 [118,662, 146,856]	-120,887 [-137,859, -104,713]	no
GLFT	159,975 [145,941, 174,942]	-146,627 [-168,554, -122,292]	no
QH_adverse	310,058 [289,083, 331,440]	359,683 [337,933, 380,284]	yes
HJB_2023	240,507 [229,928, 250,898]	56,063 [42,995, 69,667]	yes
HJB	268,618 [261,787, 275,030]	233,438 [224,310, 242,998]	yes
HJB_tox	270,060 [264,928, 275,740]	246,257 [236,938, 256,311]	yes
HJB_nwz	260,155 [254,444, 265,327]	232,384 [223,273, 241,782]	yes

Sensitivity

quoter	latency 0 ms	50 ms	200 ms	1000 ms
HJB_2023	485,586	497,850	501,347	451,308
HJB	526,806	538,120	528,184	519,747
HJB_tox	543,621	542,430	539,946	531,304

quoter	true toxicity x0	x0.5	x1	x2	x4
HJB_2023	615,182	550,384	485,586	355,990	96,799
HJB	549,800	538,303	526,806	503,812	457,823
HJB_tox	559,486	548,222	543,621	519,242	519,824

Last look

hold (ms)	tolerance (pips)	P&L $ (95% CI)	rejection rate	10 s mark-out of accepted trades (client
0	none	526,806 [516,738, 537,230]	0.0%	+0.009
0	0.5	526,806 [516,738, 537,230]	0.0%	+0.009
0	0.2	526,806 [516,738, 537,230]	0.0%	+0.009
0	0.1	526,806 [516,738, 537,230]	0.0%	+0.009
100	0.2	530,924 [517,750, 546,412]	1.5%	+0.011
200	0.2	512,533 [499,932, 522,030]	3.1%	+0.012

Tests

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TESTS PASSED: 39/39 checks
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